ScanWise

Core Clinical Module Overview

1️⃣ Multimodal Agentic Ingestion

🔎 Module Objective

Serve as the secure, intelligent entry layer for all clinical data streams before AI reasoning.

🏗️ Functional Capabilities

AI Modality Detection
Automatically identifies data type:

- MRI
- CT
- X-ray
- Laboratory Reports
- Clinical Notes

Ensures correct downstream processing pipeline selection.

Automated Anatomical Localization
AI detects anatomical region within imaging studies to:

- Pre-structure clinical tensors
- Improve contextual diagnostic reasoning
- Reduce inference ambiguity

PHI / PII Stripping (Pre-Cloud Layer)

- Local anonymization engine
- Removes identifiers before external AI processing
- HIPAA-aligned privacy-first design

PACS-Ready DICOM Compatibility

- Native DICOM parsing
- Interoperable with hospital PACS systems
- Supports imaging metadata extraction

OCR for Unstructured Documents

- Extracts text from scanned lab reports
- Converts handwritten/printed notes into structured format
- Prepares data for multimodal correlation

📦 Output Layer

Structured, anonymized clinical tensors
Ready for AI reasoning, correlation, and triage.

System Architecture & Clinical Value

🔐 Secure Data Flow Architecture

Clinical Source
→ Local Agentic Ingestion Layer
→ PHI Removal
→ Tensor Structuring
→ AI Reasoning Engine

No raw patient-identifiable data leaves the hospital environment.

🧠 Why “Agentic” Ingestion?

Unlike static ETL systems, this module:

- Dynamically selects preprocessing pipelines
- Adapts based on modality confidence
- Performs autonomous validation checks
- Flags corrupted or incomplete inputs

📊 Clinical Impact

Capability	Operational Benefit
Automated Modality Detection	Reduces manual classification errors
Anatomical Localization	Improves diagnostic confidence
PHI Stripping	Regulatory compliance
DICOM Compatibility	Seamless hospital integration
OCR Structuring	Enables multimodal AI reasoning

🎯 Strategic Advantage

This module transforms fragmented medical inputs into:

✔ Structured
✔ Secure
✔ Context-aware
✔ AI-ready clinical intelligence

It forms the foundation of the entire ScanWise diagnostic stack.

2️⃣ Clinical Reasoning & Analysis Module

🧠 Cross-Modal Diagnostic Intelligence Engine

🎯 Module Objective

Deliver high-confidence diagnostic hypotheses by correlating imaging, laboratory data, and clinical context through a structured dual-inference AI pipeline.

🔁 Dual Inference Architecture

⚡ Tier 1: Gemini 3 Flash

Purpose: Rapid preprocessing & structuring

- OCR validation
- Clinical summarization
- Lab value normalization
- Key symptom extraction
- Context compression

Provides fast, structured inputs for deeper reasoning.

🧠 Tier 2: Gemini 3 Pro

Purpose: Advanced clinical modeling

- Differential diagnosis generation
- Bayesian-style probability weighting
- Multi-factor correlation analysis
- Risk stratification modeling

Performs high-level clinical reasoning across modalities.

🔬 Imaging–Lab Correlation Engine

- Cross-maps abnormal imaging features with lab deviations
- Detects hidden co-morbid patterns
- Identifies diagnostic contradictions
- Strengthens hypothesis confidence scoring

📦 Output Layer

High-confidence, structured diagnostic hypotheses
Ranked, explainable, and clinically traceable.

Explainability, Visualization & Clinical Utility

🔎 ROI Heatmapping

- Lesion localization overlays
- Visual attention mapping
- Region-of-interest (ROI) confidence scoring
- Radiologist-aligned visual validation

Improves interpretability and trust.

🌳 Explainable Reasoning Trees

- Transparent diagnostic pathway breakdown
- Step-wise correlation logic
- Evidence-backed inference nodes
- Clear justification for each hypothesis

Supports regulatory compliance and audit readiness.

🎙️ Live Voice-Enabled Peer Review Assistant

- Real-time conversational analysis
- Interactive diagnostic clarification
- Case discussion simulation
- Multimodal data referencing during review

Designed for radiologists and clinicians in workflow.

📊 Clinical Impact

Capability	Clinical Benefit
Dual AI Pipeline	Speed + depth of reasoning
Cross-Modal Correlation	Reduced missed diagnoses
Heatmapping	Visual diagnostic confidence
Explainability Trees	Regulatory & medico-legal clarity
Voice Assistant	Faster collaborative review

🎯 Strategic Advantage

This module transforms AI from a black-box predictor into:

✔ A transparent reasoning partner
✔ A clinically aligned decision-support system
✔ A peer-review augmentation tool

It serves as the cognitive core of the ScanWise ecosystem.

Operational Triage Dashboard

3️⃣ Mission-Control Layer for Hospital Operations

🎯 Module Objective

Provide real-time, AI-driven operational intelligence to optimize triage prioritization, imaging workflows, and resource allocation.

📊 Predictive Workload Forecasting

- AI models predict case inflow trends
- Peak-hour detection
- Emergency surge anticipation
- Modality-specific demand projection

Enables proactive staffing and equipment readiness.

⏱️ Queue Velocity Tracking

- Live queue depth monitoring
- Average turnaround time (TAT) analysis
- Case aging alerts
- Bottleneck identification

Ensures critical cases are escalated promptly.

🗺️ Regional Trauma Mapping

- Geo-cluster visualization of trauma incidents
- Emergency case density heatmaps
- Seasonal / event-based surge prediction
- Referral pattern analysis

Supports system-level emergency preparedness.

📦 Output Layer

Data-driven triage prioritization recommendations
Optimized workflow routing + real-time operational insights.

Analytics Intelligence & Executive Visibility

📈 Modality Distribution Analytics

- MRI vs CT vs X-ray workload ratio
- Equipment utilization tracking
- Idle-time detection
- Maintenance risk forecasting

Improves capital efficiency and throughput.

🧾 AI-Generated Executive Clinical Briefings

- Automated daily operational summary
- High-risk case alerts
- Resource stress indicators
- Outcome prediction snapshots

Designed for hospital administrators and clinical directors.

🔄 Intelligent Prioritization Engine

AI dynamically ranks cases based on:

- Clinical severity score
- Time sensitivity
- Diagnostic uncertainty level
- Resource availability

📊 Operational Impact

Capability	Operational Benefit
Forecasting	Reduced emergency overload
Queue Tracking	Faster critical case resolution
Trauma Mapping	Regional preparedness
Utilization Analytics	Equipment optimization
Executive Briefings	Strategic decision support

🎯 Strategic Advantage

This module converts hospital operations into:

✔ Predictive
✔ Data-driven
✔ Proactively managed
✔ Resource-optimized

It acts as the command center of the ScanWise ecosystem.

Transplant Intelligence Hub

4️⃣ High-Stakes Organ Allocation Support Module

🎯 Module Objective

Augment transplant decision-making with compliance-aligned, data-driven intelligence for equitable and time-sensitive organ allocation.

🧮 Automated Clinical Scoring Engine

Supports real-time computation of:

- **MELD (Model for End-Stage Liver Disease)**
- **EPTS (Estimated Post-Transplant Survival)**
- **LAS (Lung Allocation Score)**

Features:

- Automatic lab ingestion
- Dynamic recalculation on new data
- Risk-weighted prioritization outputs

Eliminates manual scoring delays and calculation errors.

⏳ Cold Ischemia Time (CIT) Monitoring

- Continuous timer from organ retrieval
- Predictive viability decay modeling
- Threshold breach alerts
- Escalation notifications

Ensures transplant within safe physiological windows.

🧬 Multi-Factor Matching Engine

AI-assisted compatibility evaluation using:

- HLA matching
- ABO blood type compatibility
- Crossmatch probability modeling
- Sensitization risk assessment

Enhances allocation precision and reduces rejection probability.

📦 Output Layer

Ranked transplant allocation recommendations
Aligned with medical urgency, compatibility, and regulatory protocols.

Logistics Intelligence & Compliance Alignment

🚚 Logistics Simulation Engine

- Air + ground route optimization
- Real-time transit time prediction
- Weather & delay sensitivity modeling
- Backup route generation

Minimizes transport-related organ viability loss.

🚨 Organ Viability Window Alerts

- Predictive countdown dashboards
- Color-coded urgency indicators
- Multi-team notification triggers

Supports rapid surgical coordination.

⚖️ Compliance & Transparency Layer

- Audit-ready decision logs
- Score traceability
- Rule-based override documentation
- Allocation fairness analytics

Designed to align with national transplant governance frameworks.

📊 Clinical & Operational Impact

Capability	Impact
Automated Scoring	Faster, error-free prioritization
CIT Monitoring	Reduced graft loss risk
AI Matching	Improved post-transplant outcomes
Logistics Simulation	Shorter transport delays
Compliance Layer	Regulatory audit readiness

🎯 Strategic Advantage

The Transplant Intelligence Hub transforms organ allocation into:

✔ Data-driven
✔ Time-optimized
✔ Clinically precise
✔ Regulation-aligned

It supports clinicians in making high-risk, life-critical decisions with enhanced clarity and confidence.

Patient-Centric Communication Module

5️⃣ Bridging Technical Diagnostics & Patient Understanding

🎯 Module Objective

Transform complex clinical findings into structured, comprehensible, and actionable communication for patients while preserving medical accuracy.

🗣️ Radiology Jargon → Plain-Language Translation

- Converts technical terminology into patient-friendly explanations
- Maintains diagnostic integrity
- Context-aware simplification (not oversimplification)
- Adjustable complexity level

Example transformation:
“Pulmonary nodular opacity with irregular margins”
→ “A small abnormal spot was found in your lung that needs further evaluation.”

📚 Health Literacy Optimization

- Reading-level adaptation
- Multilingual capability
- Cultural sensitivity adjustments
- Risk communication clarity

Ensures comprehension across diverse patient populations.

🧾 Structured “What’s Next” Guidance

Automatically generates:

- Recommended follow-up steps
- Additional tests required
- Urgency classification
- Lifestyle or precautionary advice

Reduces ambiguity after diagnosis delivery.

📦 Output Layer

Clear, structured patient reports
Aligned with clinician-approved diagnostic conclusions.

Trust, Transparency & Secure Communication

🔐 Secure Patient–Clinician Communication Bridge

- Encrypted messaging interface
- Structured query submission
- Clarification request system
- Case-referenced discussions

Maintains privacy while enabling engagement.

🌉 Bridging the Communication Gap

This module:

- Reduces anxiety caused by technical reports
- Improves treatment adherence
- Minimizes follow-up confusion
- Supports shared decision-making

📊 Measurable Impact

Capability	Outcome
Plain-Language Translation	Improved patient understanding
Literacy Optimization	Broader accessibility
Structured Next Steps	Reduced follow-up delays
Secure Communication	Increased trust & engagement

🎯 Strategic Advantage

The Patient-Centric Communication Module ensures that AI-driven diagnostics are:

✔ Understandable
✔ Actionable
✔ Secure
✔ Trust-enhancing

It completes the ScanWise ecosystem by extending intelligence beyond clinicians to the patient experience.

EHR Bridge (FHIR R4)

6️⃣ Interoperability Backbone of ScanWise AI

🎯 Module Objective

Enable seamless, standards-compliant integration between ScanWise AI outputs and major Electronic Medical Record (EMR/EHR) systems using FHIR R4 protocols.

🔁 Bidirectional FHIR R4 Synchronization

- Real-time data push & pull
- Event-driven resource updates
- Subscription-based change tracking
- Conflict resolution handling

Ensures AI insights integrate directly into clinician workflows.

🧠 AI-Tensor to Clinical Resource Mapping

Structured AI outputs are mapped to:

- **Problem List**
- **Assessment & Plan**
- **Observations**
- **Procedures**

Preserves semantic alignment with clinical documentation standards.

📐 Data Normalization Engine

Unit Normalization

- Automatic metric ↔ imperial conversion
- Lab value standard harmonization
- Cross-system compatibility

ISO-8601 Timestamp Standardization

- Time-zone aware formatting
- Consistent longitudinal record tracking
- Inter-system temporal alignment

📦 Output Layer

FHIR-compliant structured entries synchronized into hospital EMR systems.

Audit Integrity & Enterprise Integration

📜 Immutable Audit Ledger

- Cryptographically signed AI outputs
- Decision trace logging
- Versioned inference tracking
- Override documentation

Ensures medico-legal defensibility and regulatory transparency.

🔗 Enterprise EMR Compatibility

Designed for integration with:

- Epic
- Cerner
- Meditech
- Other FHIR R4-compliant systems

Supports scalable hospital and multi-center deployment.

🔄 Secure Data Exchange Architecture

Hospital EHR
↔ FHIR Gateway
↔ AI Tensor Mapping Layer
↔ ScanWise Reasoning Engine

All exchanges encrypted and access-controlled.

📊 Clinical & Operational Impact

Capability	Outcome
FHIR R4 Sync	Native EMR integration
Resource Mapping	Structured AI documentation
Unit Normalization	Cross-system consistency
ISO Standardization	Reliable longitudinal records
Audit Ledger	Compliance & traceability

🎯 Strategic Advantage

The EHR Bridge transforms ScanWise from an external AI tool into:

✔ A fully embedded clinical intelligence layer
✔ EMR-native decision support
✔ Audit-ready and regulation-aligned infrastructure

It ensures AI insights become actionable, documented, and interoperable within real-world healthcare systems.